

Henrique dos Santos Teixeira

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EXPERIENCE

Web Application Developer *Tuna de Ciências do Porto* 02/2025 – now | Porto, Portugal

- Co-developed and maintain a full-stack web application to manage events, members, instruments and organizational workflow for a student musical group, automating previously manual workflows.
- Informally act as technical lead, managing a team of 8 student contributors, handling onboarding, development workflow and self-hosting infrastructure with Cloudflare Tunnel and JWT authentication.
- Built with React (Vite), Go and MySQL, containerized with Docker Compose.

Researcher - AIMaCoV: ONNX Graph Mapping to RISC-V CGRA *INESC TEC* 02/2026 – 07/2026 | Porto, Portugal

- Proactively advocated to lead a new research initiative, mentoring Deniz Lopes Gunes (undergraduate intern) on pipeline management and algorithmic strategy.
- Guided the design of a GNN-guided DFG partitioning flow for mapping ONNX computation graphs onto a RISC-V spatial CGRA, learning optimal partitions from SAT-generated ground truth.
- Mentored work achieved 33% edge-cut reduction over the unguided baseline, 98% mapping success vs. 15% for greedy list scheduling and 50× speedup over exact SAT (~ 0.25 s/graph).

Internship - LNS on a RISC-V SoC via High Level Synthesis *INESC TEC* 02/2026 – 06/2026 | Porto, Portugal

- Extended a prior LNSU implementation by adding comparison operations and FP/BF ↔ LNS conversion functions, approximated via greedy-generated linear splines with analytically derived error bounds.
- Built a header-only C++ LNS emulation library (lnssim.hpp) with XF and XMB spline backends, supporting lns8 Q4.3 and lns16 Q8.7 and a matching BF emulator (bfloatsim.hpp) for direct benchmarking.
- Validated lns16 Q8.7 for Llama 2 inference: direct fp32 → lns16 weight conversion without retraining produced coherent text on TinyStories 15M and 42M, with hybrid fp32 accumulation isolating LNS to matrix multiplications.

Researcher - AIQ-Ready: Accelerator for Compute-Heavy Nodes in ONNX Models *INESC TEC* 09/2025 – 01/2026 | Porto, Portugal

- Assessed viability of converting ONNX models to C via ONNX2C and profiled compute-heavy nodes in the resulting output.
- Selected bottleneck nodes for hardware acceleration on XHeep/RISC-V and evaluated combined SW + accelerated results.

Internship - AI Acceleration on a RISC-V SoC via High Level Synthesis *INESC TEC* 06/2025 – 07/2025 | Porto, Portugal

- Best CTM Internship — Computer Science and Engineering winner.
- Deployed a Logarithmic Number System Unit to a RISC-V CPU via HLS in C++, with addition/subtraction approximations using only 192 bytes of lookup tables combined.
- Integrated into the RISC-V pipeline and tested on FPGA at 125 MHz.

Academic Internship - Lambda Calculus Interpreter *FCUP* 02/2025 – 06/2025 | Porto, Portugal

- Implemented a C interpreter for Lambda Calculus compiled to SK Combinators, combining both theories to build translating and reducing functions over a custom program tree.

PAPERS

Logarithmic Number System Extension on an HLS-Based RISC-V SoC *INESC TEC (Under Review)* 2026

- 2nd author; 45% fewer LUTs, 58% fewer DSPs, 29–36% higher clock frequencies vs. IEEE-754 FPU; 95.99% MNIST accuracy without retraining.

GNN-Guided Graph Partitioning for Spatial CGRA Mapping *INESC TEC (Under Review — INFORUM 2026)* 2026

- Co-author; 33% edge-cut reduction, 98% mapping success, 50× SAT speedup.

PROJECTS

Lightweight Cloud Job Platform on GCP

- CLI-driven platform on GCP submitting shell commands as jobs on auto-provisioned GCE worker VMs, with a FastAPI backend, Redis queue, autoscaler daemon (scale-to-zero), JWT auth, rate limiting and IAP tunneling.

LNS Library

- C++ library implementing 8-bit and 16-bit Logarithmic Number System arithmetic,

with exact multiply/divide via exponent addition, spline-based LUT approximation for add/sub and both a hardware compiler header (custom RISC-V instructions) and a software simulation API for benchmarking against float32 and bfloat16.

RISC-V Assembler with Custom Instructions

- Assembler for RISC-V supporting 32IM instructions and custom LNSU instructions from INESC TEC internship.

Kotlin Compiler

- Compiler for a subset of Kotlin (statements, expressions, enums, functions) targeting MIPS assembly.

Memory Allocator Library

- Custom C library implementing an Arena (O(1) linear) and a Pool (best-fit, O(F)) allocator with coalescing, realloc and strdup support; used in the Kotlin Compiler and SK Interpreter.

EDUCATION

Licenciatura de Inteligência Artificial e Ciência de Dados *FCUP* 09/2023 – 06/2026 | Porto, Portugal

- 2× winner of “Bolsas PRR «Impulso Jovens STEAM» - Distinção Académica” scholarship.
- Relevant Courses: **Compilers** (Extra.), **AI**, **Machine Learning I & II**, **Data Structures**, **Algorithms**

TOOLS

Systems/Hardware: RISC-V, HLS, FPGAs, Docker
Infrastructure/Cloud: GCP, Redis, FastAPI, Cloudflare Tunnels, JWT
ML/Data: PyTorch, Pandas, Numpy

PROGRAMMING LANGUAGES

